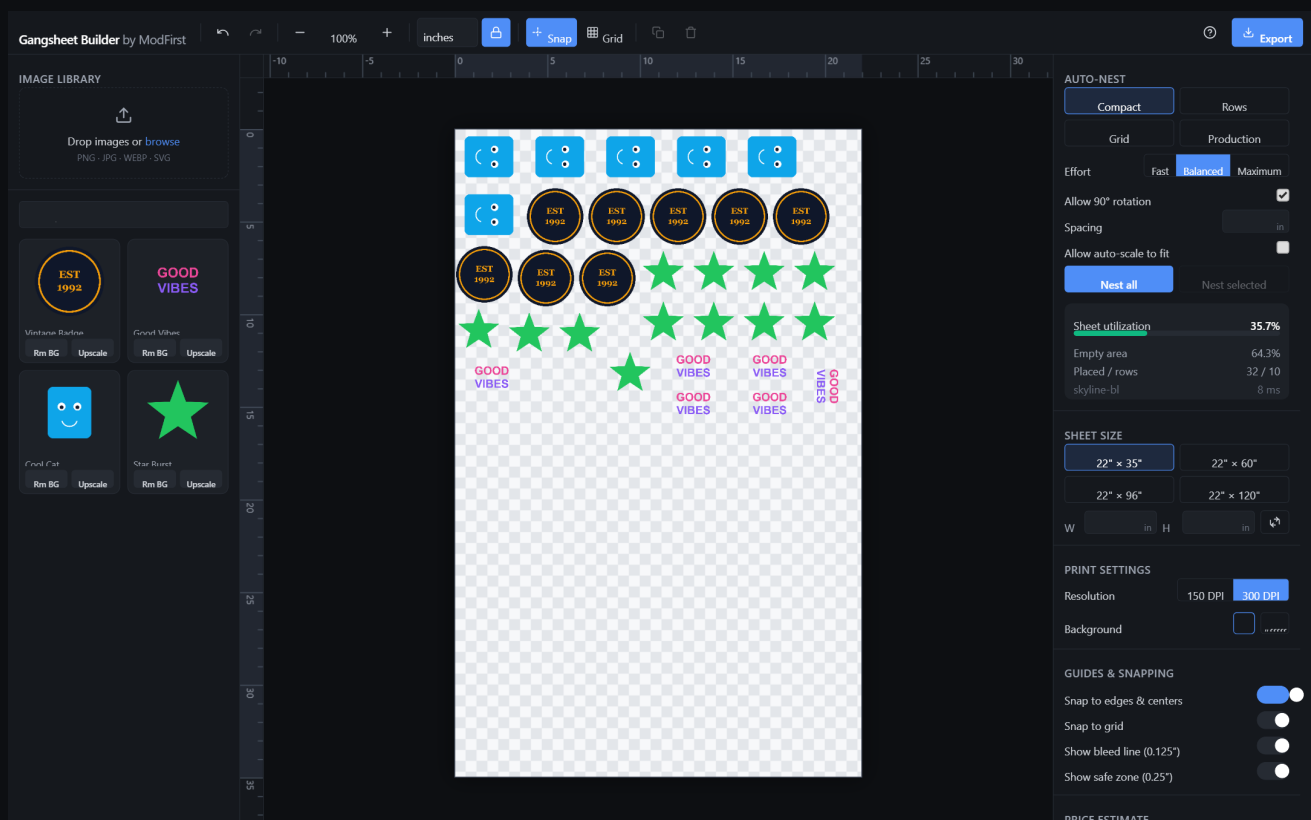


Gangsheet Builder

by ModFirst

Complete User Guide

Design, nest, and export print-ready DTF gang sheets — right in your browser.



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1 · Welcome

Gangsheet Builder by ModFirst is a browser-based tool for laying out **DTF (Direct-to-Film) gang sheets** — single print sheets packed with many designs. It is built for print shops and creators who need to fit as much artwork as possible onto a sheet, with zero wasted film and print-ready output.

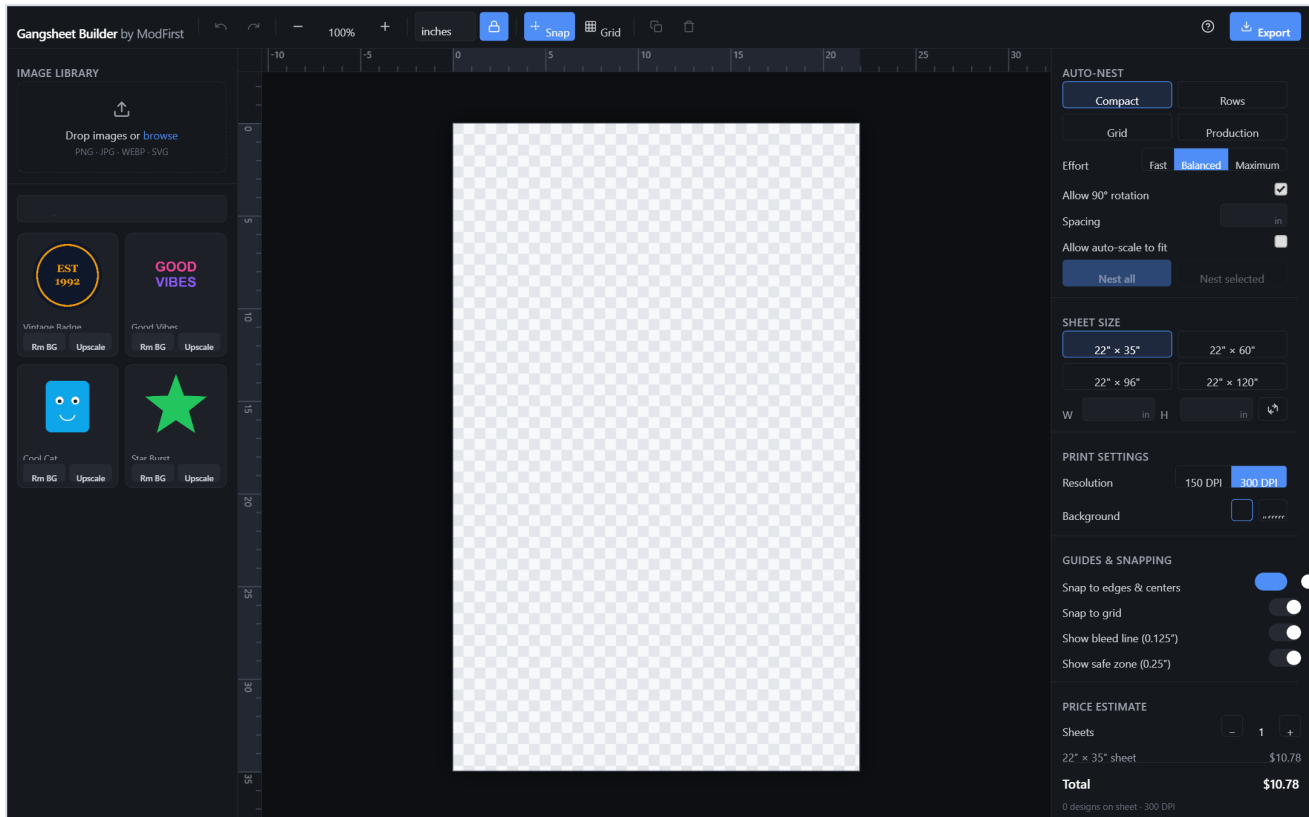
What you can do

- Upload PNG, JPG, WEBP, or SVG artwork and set the exact physical print size of each design.
- Pack designs automatically with a smart nesting engine that minimises wasted sheet area.
- Generate a full sheet in one click with **Auto Build**, or flood the free space with **Auto Fill**.
- Clean up artwork with one-click **background removal** and **upscaling**.
- Catch problems early with live overlap detection and low-resolution warnings.
- Export pixel-accurate, transparent **PNG** or print-ready **PDF** at 150, 300, or 600 DPI.

Everything is measured in real, physical units (inches), so what you arrange on screen is exactly what prints.

2 · The Workspace

The builder is split into four areas. Open it from the home screen by clicking **Open the builder**.



The builder workspace: image library (left), canvas with rulers (centre), tools panel (right).

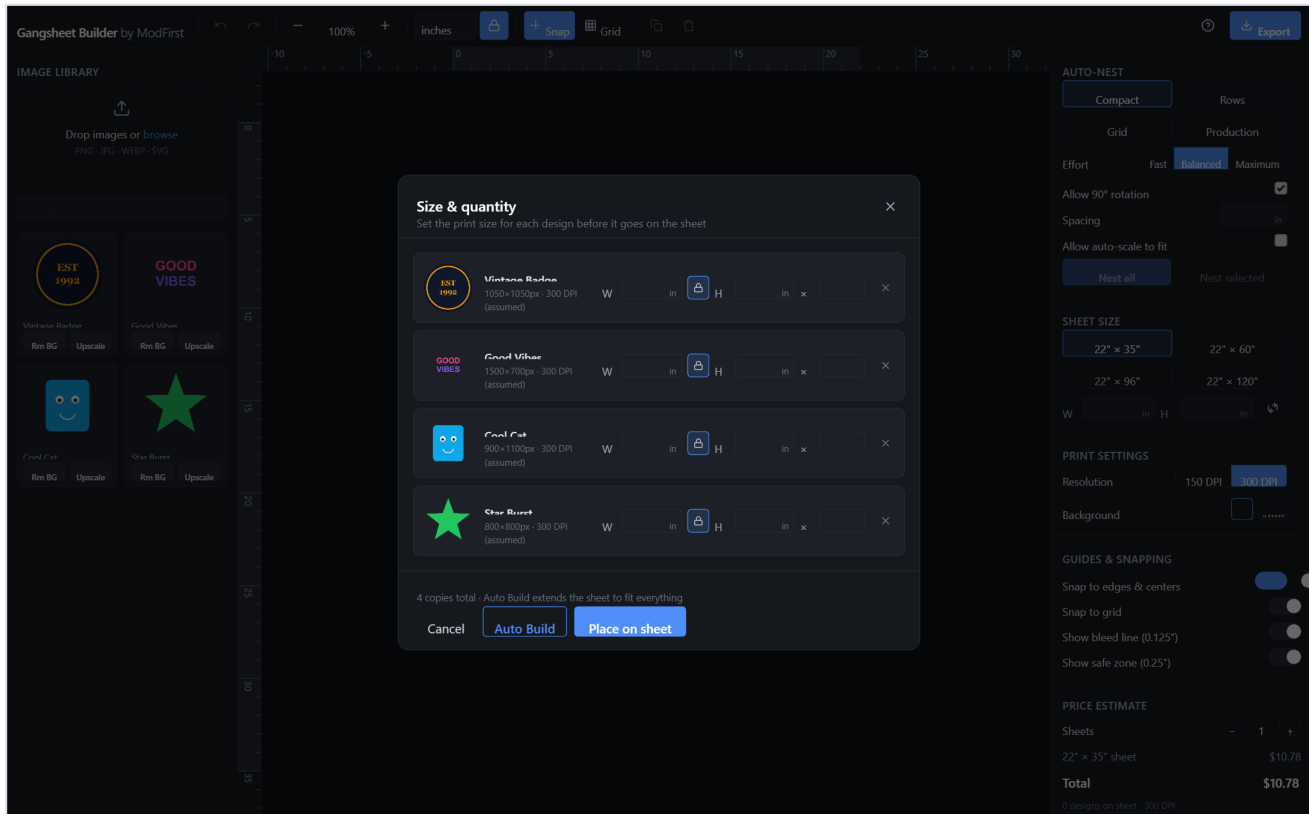
Layout at a glance

- **Top toolbar** — undo/redo, zoom, measurement units, snapping toggles, duplicate/delete, and the **Export** button.
- **Image Library (left)** — your uploaded designs, with a drag-and-drop upload zone and per-image tools.
- **Canvas (centre)** — the gang sheet shown at true scale with rulers, a checkerboard for transparency, and pan/zoom.
- **Tools panel (right)** — Auto-Nest, sheet size and print settings, guides & snapping, and a live price estimate.

Pan the canvas by holding **Space** and dragging (or middle-mouse drag); zoom with the mouse wheel. Press **?** at any time to see all keyboard shortcuts.

3 · Adding Images & Setting Size

Drag image files onto the upload zone (or click **browse**). After uploading, the **Size & quantity** window opens so you can set the real print dimensions before anything lands on the sheet.



The **Size & quantity** window — set width, height, and number of copies for each design before placing.

How sizing works

- The builder reads the **DPI from the file** when available and shows the resulting print size. If the file has no DPI, it assumes **300 DPI** (the DTF standard).
- Type a new **width** or **height** in inches. With the **lock** enabled, the aspect ratio is preserved automatically.
- Set a **quantity** to place several copies at once.
- A red warning appears if a design would print below **150 DPI** at the chosen size.

Then choose how to place them:

- **Place on sheet** — nests the copies onto the current sheet, around anything already there.
- **Auto Build** — builds a complete sheet, extending its length to fit everything (see next section).

Tip: dragging a file directly onto the canvas skips this window and drops the image at its natural size where you release it.

4 · Auto Build & Auto Fill

Auto Build

Auto Build turns a batch of designs into a finished sheet in one step. It keeps the sheet **width fixed** and extends the **height** only as far as needed to fit every copy, packing them tightly with your spacing rules.

1. Upload one or more designs and set each one's size and quantity.
2. Click **Auto Build** in the Size & quantity window.
3. The sheet resizes and all copies are nested automatically — as a single undo step.

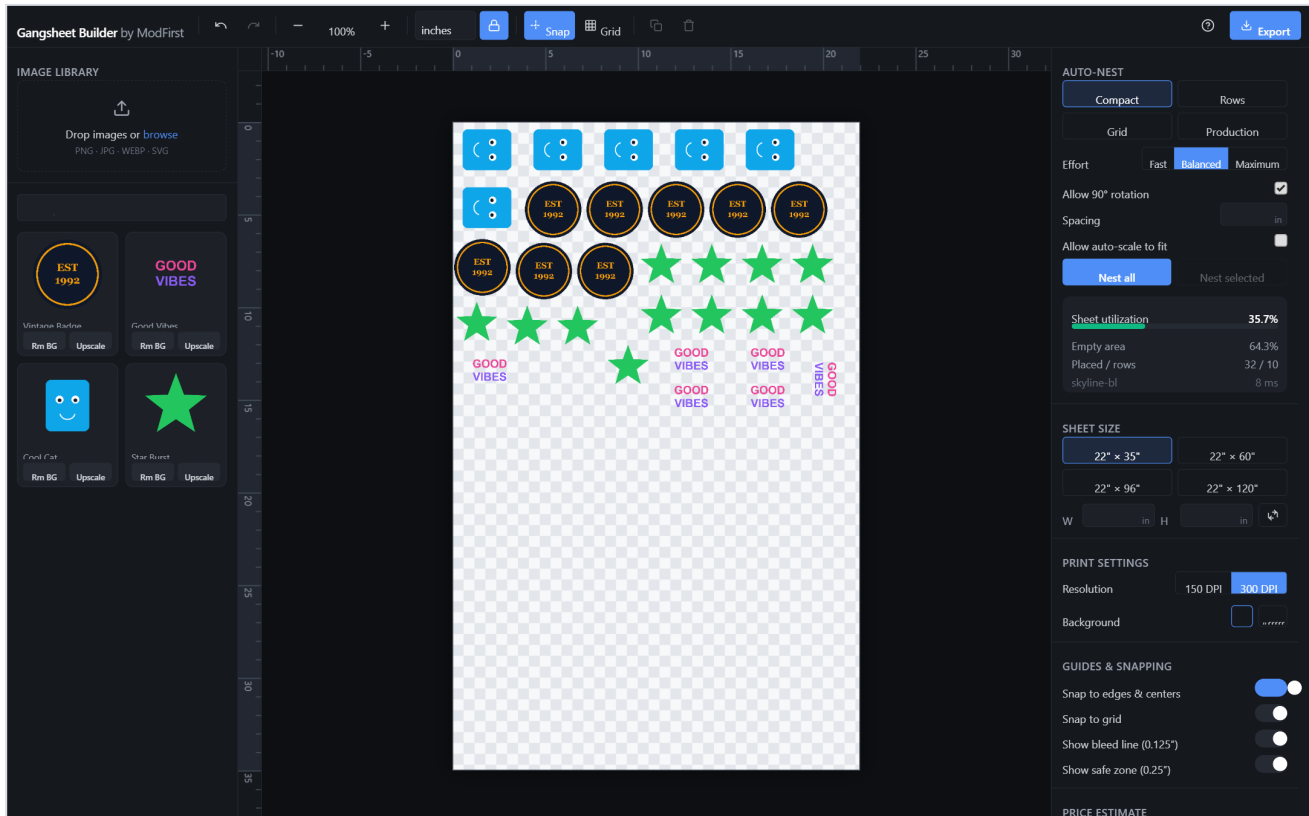
Auto Fill

Auto Fill takes one selected design and floods the remaining free area of the current sheet with as many copies as will fit, respecting spacing. Select a design on the canvas, then click **Auto Fill sheet** in the properties panel.

- Copies inherit the look of the selected design (rotation, flips, opacity).
- Existing artwork is left untouched — only the empty space is filled.
- Both actions are fully covered by **Undo** (Ctrl+Z) and **Redo**.

5 · Smart Auto-Nest

The nesting engine packs designs to maximise sheet utilisation and minimise wasted film. Use **Nest all** to repack the whole sheet, or **Nest selected** to pack only the chosen designs around everything else.



A nested sheet. The Auto-Nest panel reports utilisation, placed count, and rows.

Arrangement modes

- **Compact** — densest free-form packing for the most efficient use of film.
- **Rows** — even rows, the friendliest layout for cutting DTF transfers apart.
- **Grid** — uniform cells, ideal for sheets of identical designs.
- **Production** — runs a multi-pass search for the best achievable utilisation.

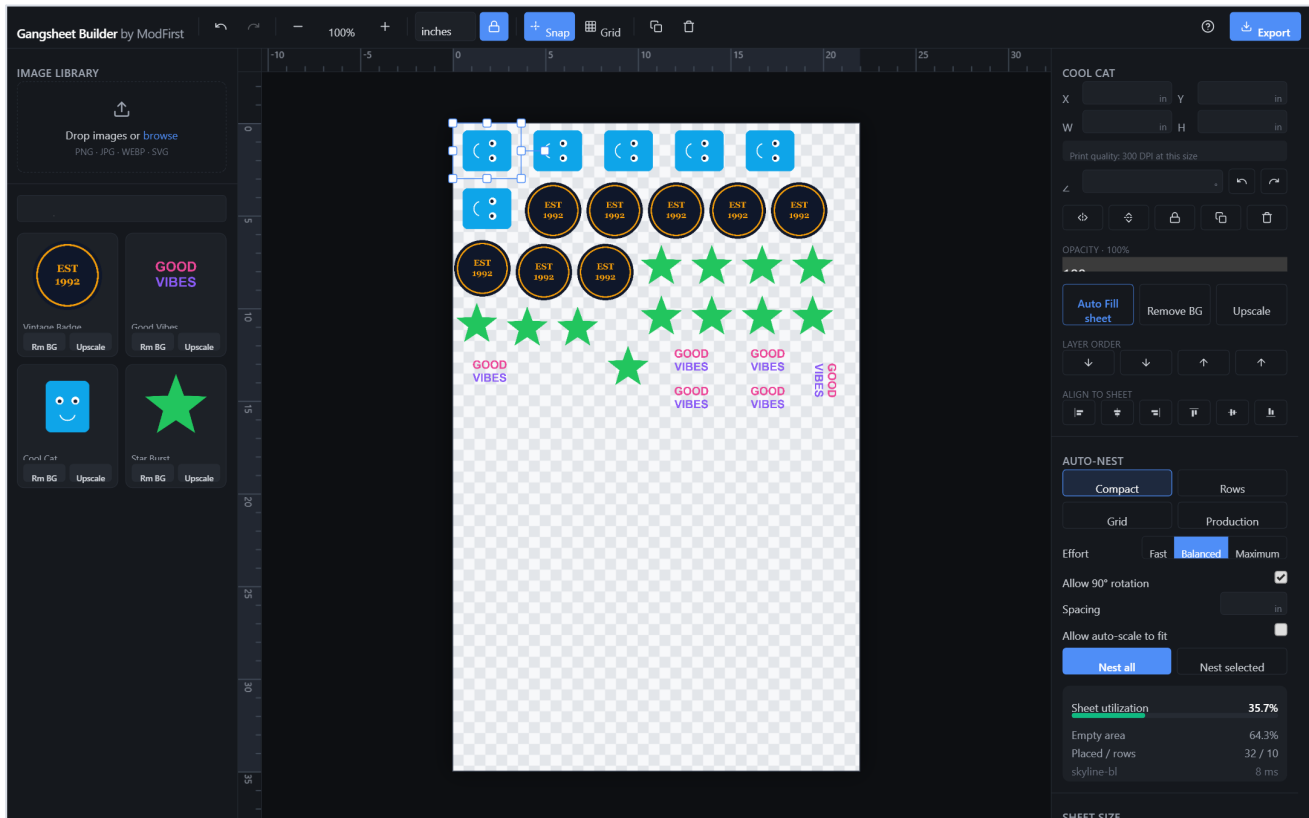
Options & results

- **Effort** (Fast / Balanced / Maximum) trades speed for tighter packing.
- **Allow 90° rotation** lets the packer turn designs when it helps them fit.
- **Spacing** sets the minimum gap between designs.
- **Allow auto-scale to fit** shrinks designs uniformly (down to your minimum) when they would otherwise overflow — with a warning if it drops below 150 DPI.
- The panel reports live **utilisation %**, empty area, placed count, and rows.

If some designs don't fit, an overflow notice offers one-click **Extend sheet** or **Auto scale** actions.

6 · Editing & Arranging Designs

Click any design to select it; its properties appear in the right panel. Drag to move, use the corner handles to resize, and the top handle to rotate. Smart guides snap designs to edges, centres, and each other.



A selected design with its properties panel: position, size, rotation, and arrangement tools.

The properties panel

- **Position & size** — exact X/Y and width/height in your chosen unit, with an aspect-ratio lock.
- **Rotation & flip** — free rotation, 90° snap buttons, and horizontal/vertical mirroring.
- **Opacity, lock, duplicate, delete** — quick per-design controls.
- **Layer order** — send to front/back or step forward/backward.
- **Align & distribute** — line designs up to the sheet or to each other, and space them evenly.
- **Auto Fill, Remove BG, Upscale** — one-click actions for the selected design.

Select multiple designs with **Shift+click** or by dragging a box around them, then move, align, or distribute the whole group at once.

7 · Background Removal & Upscaling

Every image in the library — and every selected design — has **Remove BG** and **Upscale** buttons that clean up artwork without leaving the builder.

Remove background

Strips the background to transparency, ideal for DTF where only the artwork should print. A spinner shows progress; when it finishes, every placed copy of that image updates automatically.

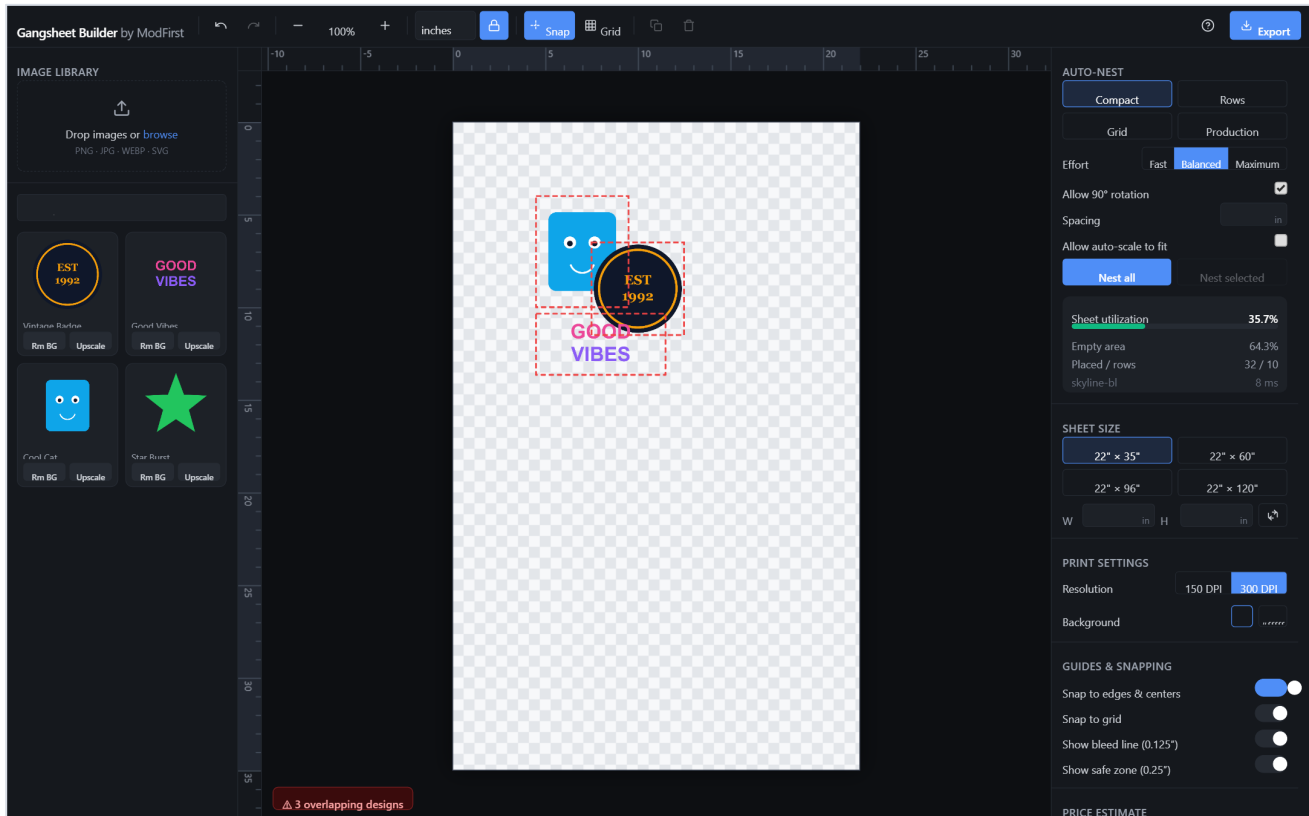
Upscale

Increases the source resolution so a design stays crisp at larger print sizes. Its physical size on the sheet is unchanged — upscaling simply raises the effective print DPI.

- Processing runs through a secure server route, so API keys are never exposed in the browser.
- Results are cached — repeating the same operation does not re-process the image.
- Each tool runs once per image; a checkmark shows when it's already been applied.
- If a tool isn't configured yet, a clear message explains what's needed.

8 - Overlap Detection

The builder checks for overlapping designs in real time — while you drag, resize, place, or nest. Overlapping designs are outlined in red and a warning appears at the bottom of the canvas.

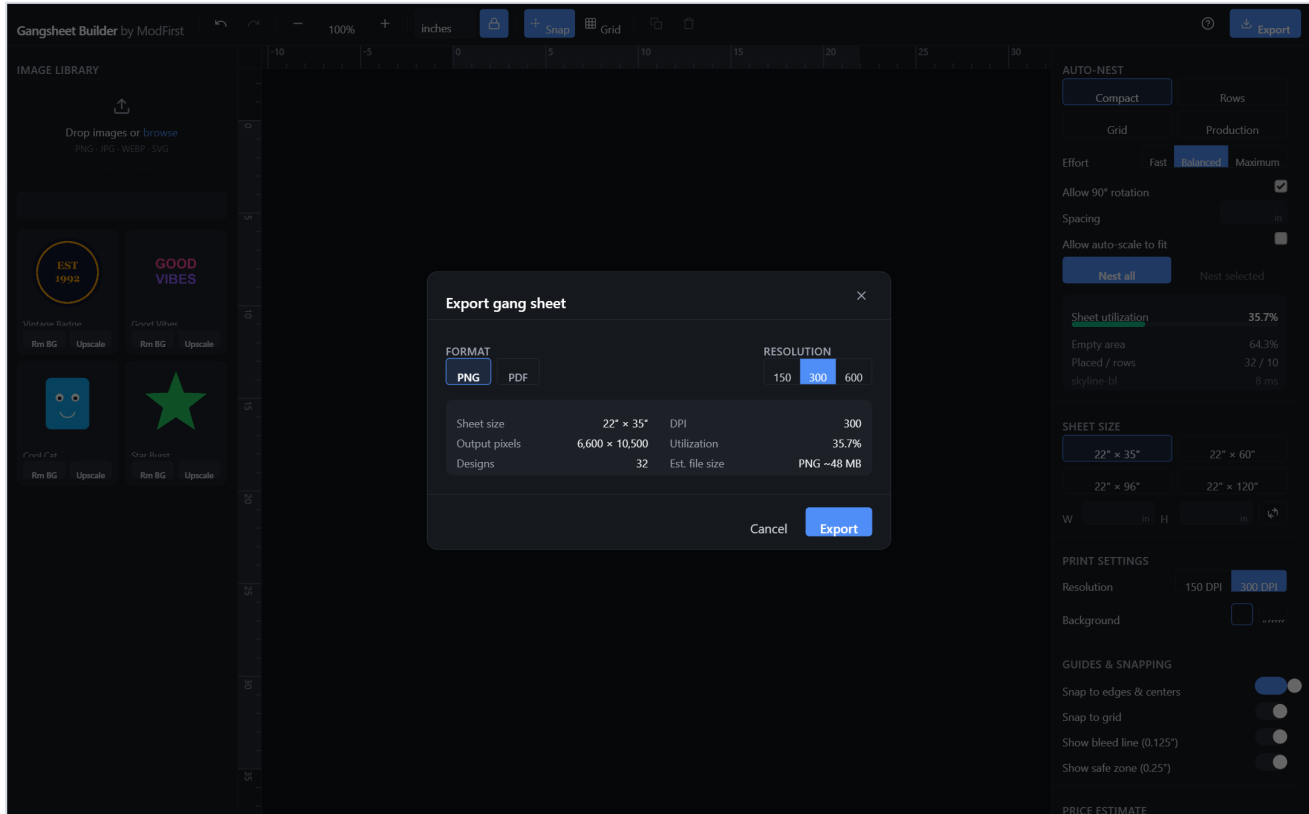


Overlapping designs are highlighted in red, with a count shown in the corner.

Overlaps are detected precisely, even for rotated designs. To resolve them, move the designs apart manually or simply run **Nest all** — the packer guarantees a layout with no overlaps.

9 · Exporting Print-Ready Files

Click **Export** in the toolbar to open the export window. It previews the output and runs pre-flight checks before producing your file.



The export window: choose format and resolution, review the summary, then export.

Formats & resolution

- **PNG** — transparent, pixel-accurate raster at the exact physical size (e.g. 22" × 35" at 300 DPI = 6,600 × 10,500 px).
- **PDF** — print-ready, with original images embedded losslessly; optional crop marks and bleed.
- **Resolution** — 150, 300, or 600 DPI. You can export several formats at once.

Pre-flight summary

Before exporting, the window shows the sheet size, output pixel dimensions, utilisation, design count, and an estimated file size. It also flags low-DPI designs, overlaps, off-sheet artwork, and missing images so you can fix them first. A progress bar tracks each export, and finished files download automatically.

10 · Keyboard Shortcuts

Press **?** in the builder to open this list at any time.

Ctrl + Z	Undo
Ctrl + Y / Ctrl + Shift + Z	Redo
Ctrl + D	Duplicate selection
Ctrl + A	Select all
Delete / Backspace	Delete selection
Escape	Clear selection
Arrow keys	Nudge 0.05"
Shift + Arrow keys	Nudge 0.5"
Shift + Click	Add / remove from selection
Mouse wheel	Zoom at cursor
Space + Drag / Middle drag	Pan canvas
Ctrl + = / Ctrl + -	Zoom in / out
Ctrl + 0	Fit sheet to view
?	Show shortcuts

11 · Tips & Troubleshooting

Get the best print results

- Keep every design at **150 DPI or higher** at its print size — watch for the red low-DPI warning and use **Upscale** if needed.
- Remove backgrounds before nesting so transparent edges pack tightly together.
- Use **Rows** mode when the sheet will be cut by hand, and **Compact** or **Production** to save the most film.
- Turn on **Show safe zone** and **Show bleed line** in Guides & Snapping to keep artwork clear of the edges.

Common questions

- **A design won't fit.** Use the overflow **Extend sheet** action, enable auto-scale, or reduce the design size.
- **Designs overlap.** Run **Nest all** for a guaranteed overlap-free layout.
- **The export is very large.** Very long sheets at 600 DPI can exceed the browser's limits — lower the DPI or shorten the sheet, and the export window will warn you in advance.
- **Something went wrong.** **Undo** (Ctrl+Z) reverses any action, including Auto Build and Auto Fill.

Gangsheet Builder by ModFirst — thank you for using the builder. Happy printing!